arXiv:2604.23295v3 [cs.CL] 24 Sep 2026

HUMAN-1: FULL-DUPLEX SPOKEN DIALOGUE MODELING FOR HINDI FROM REAL-WORLD TWO-CHANNEL CONVERSATIONS

Bhaskar Singh, Manas Dhir, Shobhit Banga, Pranav Sharma

VoiceArena

ABSTRACT

Full-duplex spoken dialogue models capture interruptions, overlaps and backchannels, yet exist almost exclusively for English. We present Human-1, the first full-duplex spoken dialogue model for Hindi. Starting from Moshi, whose tokenizer does not cover Devanagari, we train a Hindi tokenizer, reinitialize vocabulary-dependent parameters while keeping the pre-trained audio pathway and a frozen codec, and train in two stages on 26,000 hours of spontaneous two-channel Hindi conversations from 14,695 speakers. Generated speech attains a naturalness score of 4.10 versus 4.55 for human speech, with 66.9% of paired comparisons ending in a tie, and its gap and pause statistics closely match real conversations. On the IndicFDB benchmark, Human-1 produces the most human-like backchannel timing of all evaluated systems, including commercial ones. These results show that native two-channel conversational data is sufficient to transfer full-duplex behavior to a new language and script.

***Index Terms*—** Full-duplex spoken dialogue, speech language models, language adaptation, Hindi, turn-taking

1. INTRODUCTION

Human conversation is inherently full-duplex: speakers overlap, backchannel and interrupt one another, behaviors that half-duplex systems, which wait for an end-of-turn before responding, cannot reproduce [1, 2, 3]. Recent end-to-end speech dialogue models address this by jointly modeling the user and system streams in parallel [4, 5, 6, 7].

This progress has been concentrated almost entirely on English. Hindi has more than 600 million speakers, and its conversations feature frequent backchannels, pervasive Hindi–English code-mixing, and turn-taking norms that differ from those of English [8, 9], yet no full-duplex model exists for it. Adapting an existing duplex architecture to Hindi raises three challenges. (i) *Tokenization:* existing duplex models use English-centric text vocabularies; Moshi's tokenizer does not cover Devanagari, so Hindi text cannot be represented in its time-aligned text stream at all. (ii) *Partial re-training:* replacing the tokenizer invalidates every vocabulary-dependent parameter, so adaptation becomes partial pre-training rather than conventional fine-tuning. (iii) *Data:* duplex models must learn from two-channel recordings of natural interaction, and resources at this scale are, to our knowledge, unavailable for any Indian language.

We address all three and make the following contributions:

- A 26,000-hour corpus of spontaneous two-party Hindi conversations from 14,695 speakers, recorded natively on separate channels, which to our knowledge is the largest purpose-built conversational speech corpus for an Indian language.
- A language-adaptation recipe for Moshi: tokenizer replacement with targeted reinitialization, a frozen codec, and two-stage training, together with an analysis of its training dynamics.
- Human-1, the first full-duplex spoken dialogue model for Hindi, evaluated with codec, perplexity, turn-taking and large-scale human evaluation, and benchmarked against commercial voice assistants on IndicFDB.

2. RELATED WORK

dGSLM [1] was among the first models to generate two-channel spoken dialogue jointly. Moshi [4] made real-time full-duplex dialogue practical by combining a 7B text LLM (Helium), a streaming neural codec (Mimi) in the lineage of [10, 11], and a hierarchical RQ-Transformer. SyncLLM [5], OmniFlatten [6] and SALMONN-omni [7] explore alternative duplex formulations. All of these systems target English or, in some cases, Chinese. Multilingual speech–text models such as SpeechGPT [12] and AudioPaLM [13] operate in a turn-based setting, and cross-lingual transfer has mainly been studied for ASR [14] and TTS [15]. Adapting a full-duplex model to a new language, especially one with a different script, has not been studied, and to our knowledge no full-duplex system exists for any Indian language.

3. HINDI CONVERSATIONAL CORPUS

Collection. We collected 26,000 hours of unscripted two-party Hindi conversations on diverse topics from 14,695 paid participants (Table 1). Each speaker was recorded on a separate channel, which yields true stereo dialogue. Moshi instead builds pseudo-stereo training data by diarizing monophonic recordings [4]; native channel separation avoids the diarization errors and overlap artifacts of that approach. Since turn-taking, backchannels and overlaps occur only in real interaction, we deliberately avoided read speech, broadcast and monologue sources. The corpus is part of an ongoing effort to collect one million hours of spontaneous Hindi conversation.

Quality control and alignment. Trained annotators screened each recording for clarity, speaker balance, transcription accuracy and conversational naturalness, and recordings below threshold were discarded. Transcripts were produced by a combination of manual and automatic transcription, and word-level timestamps were obtained with WhisperX [16]. Text tokens are placed at their aligned 12.5 Hz frames, and PAD tokens fill frames without text. Each training sample spans 2,048 frames (≈2.7 min). The PAD ratio is ≈75%, higher than the ≈65% reported for English [4]. We hypothesize that this reflects the Hindi-specific vocabulary, which needs fewer text tokens per second of speech.

Fine-tuning and test splits. From the corpus we manually curate 1,000 hours with clear articulation, low background noise, balanced participation and natural prosody. About 10 hours are held out for validation and checkpoint selection, and the remaining ≈990 hours are used for fine-tuning. All evaluations use a separately recorded set of conversations, disjoint from the training corpus.

Table 1. Hindi conversational corpus.

Total duration	26,000 h (1,000 h curated subset)
Unique speakers	14,695
Style	Spontaneous, unscripted, two-party
Channels	Stereo, one speaker per channel
Quality control	Annotator screening + manual checks

4. METHOD

4.1. Base architecture

Moshi [4] has two components. **Mimi** encodes 24 kHz audio into 8 residual codebooks at 12.5 Hz (1.1 kbps); the first codebook is distilled to carry semantic content and the remaining seven carry acoustic detail. The **RQ-Transformer** consists of a 7B-parameter *Temporal Transformer*, which at each step s consumes 17 parallel streams (1 text, 8 system audio, 8 user audio) and outputs a context vector z_s from which a text token is predicted through a linear head, and a small *Depth Transformer*, which autoregressively predicts the 16 audio tokens of step s conditioned on z_s. Acoustic tokens are delayed by one step relative to semantic tokens to stabilize generation.

We keep Mimi **frozen** throughout, because it reconstructs Hindi with adequate quality without retraining (Sec. 5.3). Only the RQ-Transformer is trained, which substantially reduces cost.

4.2. Tokenizer replacement and reinitialization

Moshi predicts a time-aligned text stream (its "inner monologue") alongside audio, so it can only be trained on a language its text tokenizer can represent. Moshi's SentencePiece [17] tokenizer was trained for English and does not cover the Devanagari script, which makes replacing it a prerequisite rather than an optimization. We train a 32,000-token SentencePiece model on a large Hindi text corpus. The vocabulary size is unchanged, but the token identities change, so we reinitialize every vocabulary-dependent parameter: the text embedding tables of both the Temporal and Depth Transformers and the output text head. All other weights, including the audio embeddings and the transformer blocks, are kept. As a result, text representations are learned from scratch while the pre-trained audio pathway is preserved, which makes the first stage *partial pre-training* rather than fine-tuning.

4.3. Two-stage training

Stage 1 (pre-training). Because the text embeddings are freshly initialized, we use Moshi's original pre-training learning rate, 3×10^{-5}, instead of the smaller rates typical of adaptation, with AdamW [18] (β_1=0.9, β_2=0.95, ϵ=10^{-5}, weight decay 0.1). Following [4], the loss on PAD tokens is down-weighted by 50% and semantic and acoustic audio tokens are weighted 100:1. We train on 8×H100 (80 GB) GPUs in bf16 with an effective batch of 64 samples (per-device batch 4, gradient accumulation 2) for one epoch (≈10k steps, ≈13 h).

Stage 2 (fine-tuning). Starting from the Stage-1 checkpoint, we fine-tune on the ≈990-hour curated set with an effective batch of 16, 50 warm-up steps and component-specific learning rates: 2×10^{-6} for the Temporal Transformer and 4×10^{-6} for the Depth Transformer. The higher Depth-Transformer rate allows faster adaptation of acoustic prediction while limiting drift in the newly learned text representations. We validate and checkpoint every 802 steps.

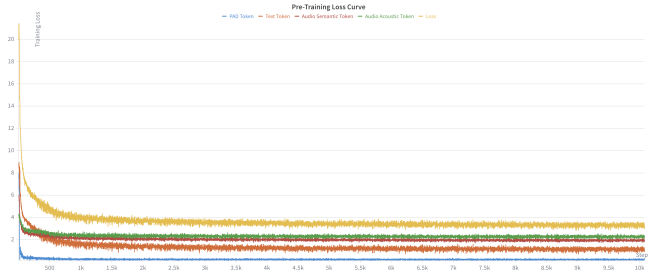

Fig. 1. Stage-1 training loss on 26,000 h of Hindi conversation. All components converge within 2k–4k steps.

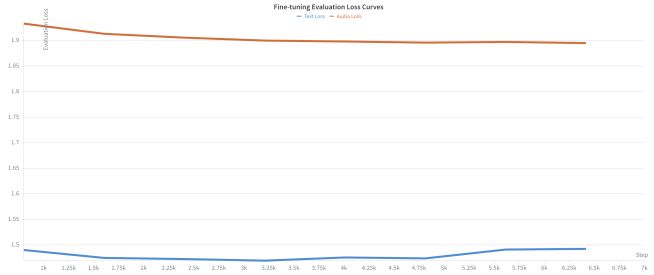

Fig. 2. Stage-2 validation loss. Text loss overfits beyond ∼4.8k steps while audio loss improves marginally. The selected checkpoint is step 4,812.

5. EXPERIMENTS AND RESULTS

5.1. Training dynamics

Pre-training. All loss components drop sharply during the first ∼2k steps (Fig. 1), and the total loss falls from ≈20 at initialization to ≈3.5, where it plateaus after ∼4k steps, i.e., before half an epoch. At convergence, text accuracy on non-PAD tokens is ≈70% and audio accuracy is 48–53% across codebooks. Even though the text embeddings and head start from random initialization, the text loss converges almost as fast as the audio losses, which indicates that the retained transformer weights transfer across scripts.

Fine-tuning. The text and audio losses behave differently on the validation set (Fig. 2). The text validation loss flattens after ∼1.6k steps and rises steadily beyond ∼4.8k steps, a clear sign of overfitting to the curated subset. The audio validation loss continues to decrease slowly with diminishing returns. We select the checkpoint at step 4,812, which minimizes total validation loss (3.370; text 1.474, audio 1.896). Training loss continues to decrease well past this point, so validation-based early stopping is essential when fine-tuning on limited curated data.

5.2. Evaluation protocol

We follow the prompted dialogue continuation protocol [1, 4]. Each held-out dialogue is split into 30 s segments: the first 10 s of both channels serve as the prompt, and the model generates the next 20 s at sampling temperature $\tau \in \{0.8, 0.9, 1.0\}$.

5.3. Codec fidelity on Hindi

To test whether Mimi, trained predominantly on English, generalizes to Hindi, we compare 654 twenty-second test segments with their Mimi re-syntheses. STOI of 0.878 ± 0.027 [19] indicates that intelligibility is largely preserved. PESQ of 2.55 ± 0.37 [20] is moderate,

Table 2. Perplexity of generated speech (Whisper-v3 transcripts scored by Sarvam-1) and turn-taking statistics (s/min).

	τ	PPL↓	IPU	Pause	Gap	Overlap
Ground truth	–	237.1	35.30	10.49	8.51	3.03
Human-1	0.8	**356.9**	23.12	9.16	6.77	1.67
	0.9	467.1	29.14	**9.24**	**8.54**	**4.30**
	1.0	640.6	**38.90**	11.67	8.10	9.68

Table 3. Human evaluation (130 raters, 2,125 paired comparisons). Rubric pass rates are rounded.

Perceptual (1–5)	Human	Human-1
Naturalness	4.55	4.10
Clarity	4.05	3.04
Preferred (tie: 66.9%)	30.0%	3.1%
Rubric pass rate (Human-1)		
Human-like interaction	85%	
Appropriateness (follows prompt)	53%	
Completion (complete reply)	42%	

which is expected at 1.1 kbps for a codec designed for low-latency streaming rather than maximal fidelity. On this basis we freeze Mimi.

5.4. Linguistic quality and turn-taking

We transcribe the generated speech with Whisper-large-v3 [21] and score the transcripts with Sarvam-1 (2B) [22], a Hindi-capable LM, reporting perplexity (PPL) as in [1, 4]. For turn-taking we apply energy-based VAD independently to each channel and compute the cumulative duration (s/min) of inter-pausal units (IPU), pauses (within-speaker silences), gaps (between-speaker silences) and overlaps [1].

Table 2 shows a trade-off controlled by temperature. PPL is lowest at τ=0.8 and rises with τ as sampling becomes more diverse. The turn-taking statistics, however, are closest to ground truth at τ=0.9: the gap duration nearly matches the reference (8.54 vs. 8.51 s/min) and the pause duration is comparable (9.24 vs. 10.49). At τ=0.8 the model is overly conservative, with less speech and little overlap, whereas at τ=1.0 overlap is three times the reference (9.68 vs. 3.03). The ground-truth PPL of 237.1 is itself high, which reflects ASR errors on spontaneous, code-mixed speech and the domain mismatch between written-text LMs and conversational transcripts. PPL is therefore best interpreted relative to this reference.

5.5. Human evaluation

In total, 130 native Hindi speakers completed 2,125 paired comparisons between human and model continuations of the same prompt. Raters scored naturalness and clarity on 5-point scales, stated a preference, and answered binary rubric questions on conversational quality (Table 3).

Human-1 reaches a naturalness score of 4.10, against 4.55 for human speech, and two thirds of the pairwise comparisons (66.9%) are ties, so listeners often cannot distinguish its speech from real conversation. Clarity (3.04 vs. 4.05) is weaker, which is consistent with the moderate PESQ of the frozen codec. The rubric results locate the main limitation in content rather than form: 85% of responses are judged human-like, but only 53% follow the prompt context appropriately and 42% form complete replies. Semantic coherence over longer contexts is therefore the main direction for improvement.

5.6. Benchmark comparison on IndicFDB

We further evaluate Human-1 on IndicFDB [23], a concurrent benchmark to which some of the present authors contributed, which extends the Full-Duplex-Bench protocol [24] to Indian languages and measures four interactive behaviors: *pause handling* (not taking the floor during a user's mid-utterance pause), *turn-taking* (responding when the user finishes), *backchanneling* (success rate, Jensen–Shannon distance (JSD) between agent and human backchannel timing, and backchannel frequency) and *user interruption* (yielding the floor, latency, and a 0–5 LLM rating of the relevance and quality of the post-interruption response, obtained after transcription and translation to English). Table 4 reports the Hindi results. Apart from Human-1, all evaluated systems are proprietary production voice assistants.

Human-1 achieves the lowest backchannel JSD of all systems (0.795), i.e., its backchannel timing is the closest to human behavior, and its backchannel frequency (0.110 vs. 0.079 for humans) is on par with the best system (0.109), while its backchannel success rate is three times that of GPT Live (45.97% vs. 15.32%). The commercial systems other than GPT Live either almost never backchannel (frequency $\leq$0.010) or do so with timing far from human (JSD $\geq$0.94), despite high success rates. This supports our central claim that training on native two-channel conversations teaches listener behavior that is otherwise difficult to obtain. In turn-taking success Human-1 is close to GPT Live (89.36% vs. 91.06%) and ahead of Grok (85.11%), and it outperforms GPT Live in pause handling (72.06% vs. 61.59%). Human-1 also yields the floor in 95.67% of user interruptions. Its main weaknesses are response latency ($\approx$2 s) and the low rating of its post-interruption responses (0.564). The same judging pipeline rates the other Hindi systems at 4.857–4.973, so transcription and translation errors are unlikely to explain this gap fully [23]. The low rating is consistent with the appropriateness and completion rubrics in Table 3: Human-1 handles the timing of interaction well, but the semantic content of its replies lags behind systems built on large proprietary text LLMs. For reference, the original English Moshi, which cannot operate in Hindi, obtains 57.46% pause handling, 71.06% turn-taking and 3.12% backchannel success on the Indian-English split of the same benchmark [23]; Human-1 exceeds all three in Hindi.

6. CONCLUSION

We presented Human-1, a full-duplex spoken dialogue model for Hindi obtained by adapting Moshi through tokenizer replacement, targeted reinitialization and two-stage training on 26,000 hours of native two-channel conversations. Mimi transfers to Hindi without retraining, so adaptation can be confined to the RQ-Transformer. The model learns realistic turn-taking directly from real conversations, without synthetic dialogue data, and on IndicFDB its backchannel timing is closer to human behavior than that of any commercial system evaluated. We also report training dynamics that can guide adaptation of Moshi to other languages. Moshi was trained on about 7M hours of speech, whereas Human-1 reaches natural full-duplex behavior with 26k hours. Scaling quality-controlled conversational data therefore appears to be the most promising route to fluent and coherent duplex dialogue for Hindi and other Indian languages.

Table 4. Hindi results on IndicFDB [23]. S.R.: success rate (%); Lat.: latency (s); JSD: Jensen–Shannon distance; Freq.: backchannels/s (closer to Human is better); Rating: interruption response rating (0–5). All systems except Human-1 are proprietary.

	Pause handling	Turn taking		Agent backchanneling			User interruption		
Model	S.R.↑	S.R.↑	Lat.↓	S.R.↑	JSD↓	Freq.↔	S.R.↑	Lat.↓	Rating↑
Human	100.00	100.00	0.259	100.00	0.000	0.079	–	–	–
GPT Live	61.59	91.06	**0.181**	15.32	0.822	**0.109**	**100.00**	**0.322**	4.857
Grok	99.05	85.11	2.556	98.18	0.986	0.004	99.67	1.628	4.960
Gemini 3.8	**100.00**	**99.15**	1.400	98.18	0.942	0.010	**100.00**	1.255	4.970
Gemini 3.1	**100.00**	98.30	2.128	**100.00**	0.990	0.003	**100.00**	1.678	**4.973**
Human-1 (ours)	72.06	89.36	2.021	45.97	**0.795**	0.110	95.67	1.573	0.564

7. REFERENCES

[1] Tu Anh Nguyen, Eugene Kharitonov, Jade Copet, Yossi Adi, Wei-Ning Hsu, Ali Elkahky, Paden Tomasello, Robin Algayres, Benoît Sagot, Abdelrahman Mohamed, and Emmanuel Dupoux, "Generative spoken dialogue language modeling," *Trans. Assoc. Comput. Linguist.*, vol. 11, pp. 250–266, 2023.

[2] Peng Wang, Songshuo Lu, Yaohua Tang, Sijie Yan, Wei Xia, and Yuanjun Xiong, "A full-duplex speech dialogue scheme based on large language model," in *Proc. Adv. Neural Inf. Process. Syst. (NeurIPS)*, 2024.

[3] Ziyang Ma, Yakun Song, Chenpeng Du, Jian Cong, Zhuo Chen, Yuping Wang, Yuxuan Wang, and Xie Chen, "Language model can listen while speaking," *arXiv preprint arXiv:2408.02622*, 2024.

[4] Alexandre Défossez, Laurent Mazaré, Manu Orsini, Amélie Royer, Patrick Pérez, Hervé Jégou, Edouard Grave, and Neil Zeghidour, "Moshi: A speech-text foundation model for real-time dialogue," *arXiv preprint arXiv:2410.00037*, 2024.

[5] Bandhav Veluri, Benjamin N. Peloquin, Bokai Yu, Hongyu Gong, and Shyamnath Gollakota, "Beyond turn-based interfaces: Synchronous LLMs as full-duplex dialogue agents," in *Proc. Conf. Empirical Methods Nat. Lang. Process. (EMNLP)*, 2024, pp. 21390–21402.

[6] Qinglin Zhang, Luyao Cheng, Chong Deng, Qian Chen, Wen Wang, Siqi Zheng, Jiaqing Liu, Hai Yu, and Chaohong Tan, "OmniFlatten: An end-to-end GPT model for seamless voice conversation," *arXiv preprint arXiv:2410.17799*, 2024.

[7] Wenyi Yu, Siyin Wang, Xiaoyu Yang, Xianzhao Chen, Xiaohai Tian, Jun Zhang, Guangzhi Sun, Lu Lu, Yuxuan Wang, and Chao Zhang, "SALMONN-omni: A codec-free LLM for full-duplex speech understanding and generation," *arXiv preprint arXiv:2411.18138*, 2024.

[8] Reiko Hayashi, "Simultaneous talk—from the perspective of floor management of English and Japanese speakers," *World Englishes*, vol. 7, no. 3, pp. 269–288, 1988.

[9] Maria Stubbe, "Are you listening? Cultural influences on the use of supportive verbal feedback in conversation," *J. Pragmatics*, vol. 29, no. 3, pp. 257–289, 1998.

[10] Neil Zeghidour, Alejandro Luebs, Ahmed Omran, Jan Skoglund, and Marco Tagliasacchi, "SoundStream: An end-to-end neural audio codec," *IEEE/ACM Trans. Audio, Speech, Lang. Process.*, vol. 30, pp. 495–507, 2022.

[11] Alexandre Défossez, Jade Copet, Gabriel Synnaeve, and Yossi Adi, "High fidelity neural audio compression," *Trans. Mach. Learn. Res.*, 2023.

[12] Dong Zhang, Shimin Li, Xin Zhang, Jun Zhan, Pengyu Wang, Yaqian Zhou, and Xipeng Qiu, "SpeechGPT: Empowering large language models with intrinsic cross-modal conversational abilities," in *Findings Assoc. Comput. Linguist. (EMNLP)*, 2023, pp. 15757–15773.

[13] Paul K. Rubenstein, Chulayuth Asawaroengchai, Duc Dung Nguyen, et al., "AudioPaLM: A large language model that can speak and listen," *arXiv preprint arXiv:2306.12925*, 2023.

[14] Alexis Conneau, Alexei Baevski, Ronan Collobert, Abdelrahman Mohamed, and Michael Auli, "Unsupervised cross-lingual representation learning for speech recognition," in *Proc. Interspeech*, 2021, pp. 2426–2430.

[15] Ziqiang Zhang, Long Zhou, Chengyi Wang, et al., "Speak foreign languages with your own voice: Cross-lingual neural codec language modeling," *arXiv preprint arXiv:2303.03926*, 2023.

[16] Max Bain, Jaesung Huh, Tengda Han, and Andrew Zisserman, "WhisperX: Time-accurate speech transcription of long-form audio," in *Proc. Interspeech*, 2023, pp. 4489–4493.

[17] Taku Kudo and John Richardson, "SentencePiece: A simple and language independent subword tokenizer and detokenizer for neural text processing," in *Proc. Conf. Empirical Methods Nat. Lang. Process. (EMNLP): Syst. Demonstrations*, 2018, pp. 66–71.

[18] Ilya Loshchilov and Frank Hutter, "Decoupled weight decay regularization," in *Proc. Int. Conf. Learn. Represent. (ICLR)*, 2019.

[19] Cees H. Taal, Richard C. Hendriks, Richard Heusdens, and Jesper Jensen, "An algorithm for intelligibility prediction of time–frequency weighted noisy speech," *IEEE Trans. Audio, Speech, Lang. Process.*, vol. 19, no. 7, pp. 2125–2136, 2011.

[20] Antony W. Rix, John G. Beerends, Michael P. Hollier, and Andries P. Hekstra, "Perceptual evaluation of speech quality (PESQ)—a new method for speech quality assessment of telephone networks and codecs," in *Proc. IEEE Int. Conf. Acoust., Speech, Signal Process. (ICASSP)*, 2001, pp. 749–752.

[21] Alec Radford, Jong Wook Kim, Tao Xu, Greg Brockman, Christine McLeavey, and Ilya Sutskever, "Robust speech recognition via large-scale weak supervision," in *Proc. Int. Conf. Mach. Learn. (ICML)*, 2023, pp. 28492–28518.

[22] Sarvam AI, "Sarvam-1: A 2B parameter language model optimized for Indian languages," `https://huggingface.co/sarvamai/sarvam-1`, 2024.

[23] Rajarshi Roy, Shobhit Banga, Jonathan Raiman, Supriya Paul, Bhaskar Singh, Manmeet Kaur, Sagar Jain, Hanuman Sidh, Pranav Sharma, Aditya Singh, Aaditya Pareek, Manas Dhir, Adi Margolin, Niket Agarwal, and Bryan Catanzaro, "IndicFDB: Benchmarking full-duplex voice agents across Indian languages," `https://3bk47sbkf7b.github.io/indicfdb/paper.pdf`, 2026.

[24] Guan-Ting Lin, Jiachen Lian, Tingle Li, Qirui Wang, Gopala Anumanchipalli, Alexander H. Liu, and Hung-yi Lee, "Full-Duplex-Bench: A benchmark to evaluate full-duplex spoken dialogue models on turn-taking capabilities," *arXiv preprint arXiv:2503.04721*, 2025.